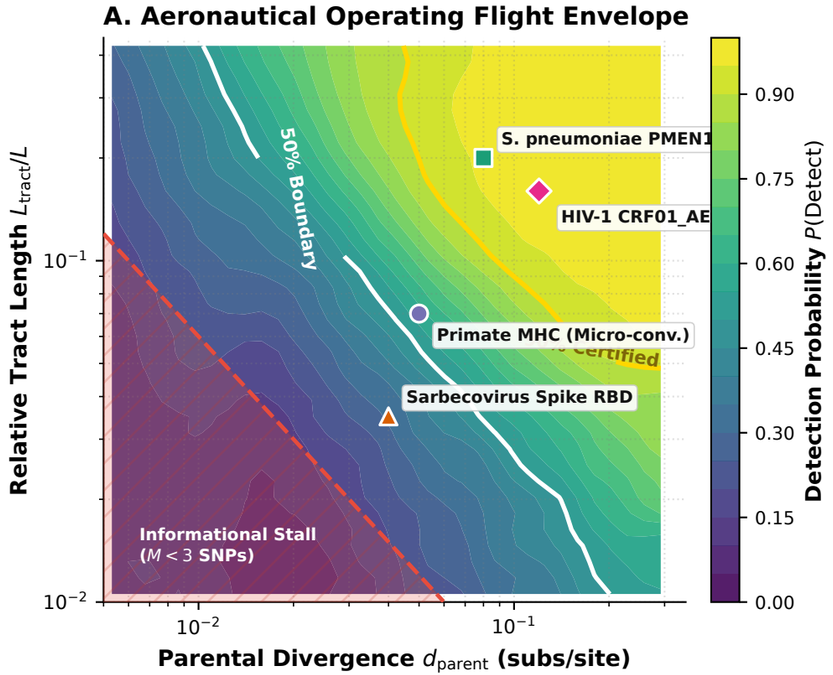
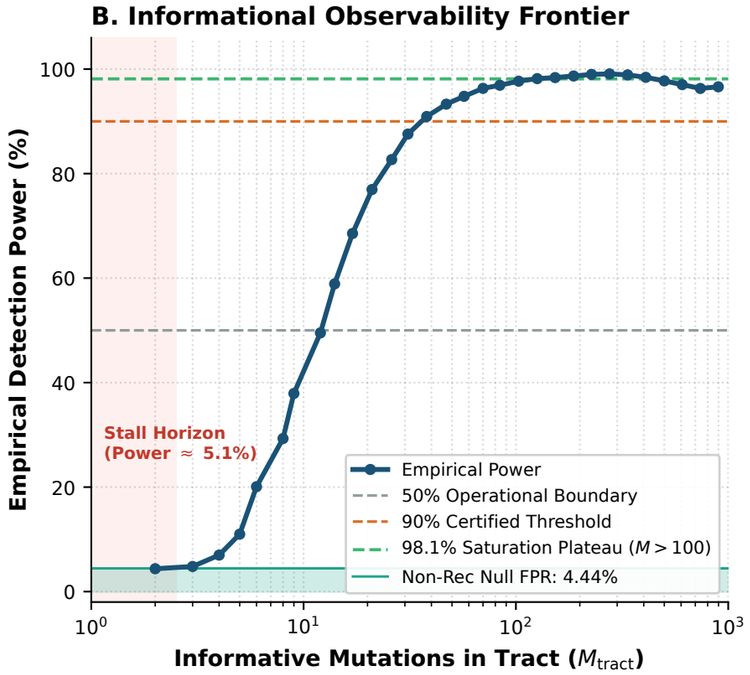
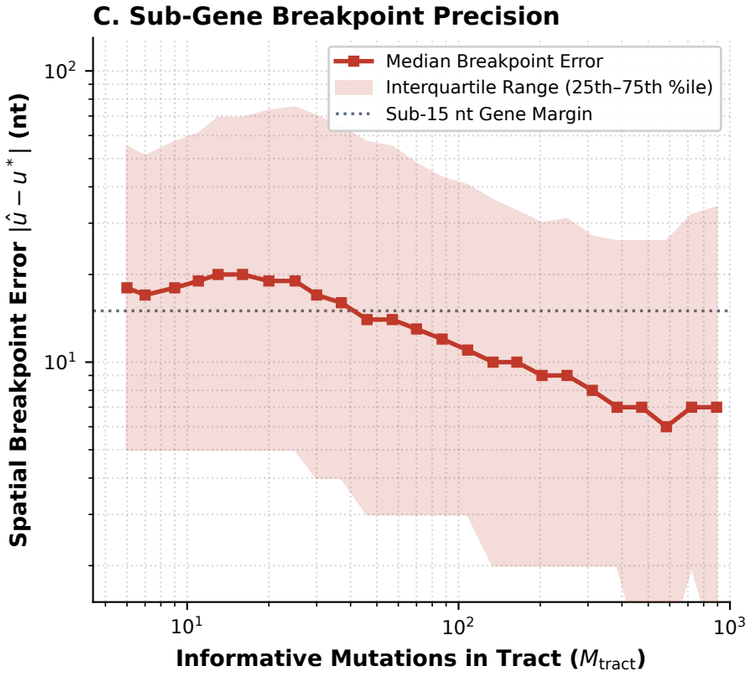

A. Aeronautical Operating Flight Envelope
S. pneumoniae PMEN1
HIV-1 CRF01_AE
Primate MHC (Micro-conv.)
Sarbecovirus Spike RBD
50% Boundary
Informational Stall
(M < 3 SNPs)
Relative Tract Length L_tract/L
Parental Divergence d_parent (subs/site)
Detection Probability P(Detect)


B. Informational Observability Frontier
Stall Horizon
(Power ≈ 5.1%)
Empirical Power
50% Operational Boundary
90% Certified Threshold
98.1% Saturation Plateau (M > 100)
Non-Rec Null FPR: 4.44%
Empirical Detection Power (%)
Informative Mutations in Tract (M_tract)


C. Sub-Gene Breakpoint Precision
Median Breakpoint Error
Interquartile Range (25th–75th %ile)
Sub-15 nt Gene Margin
Spatial Breakpoint Error |û − u*| (nt)
Informative Mutations in Tract (M_tract)